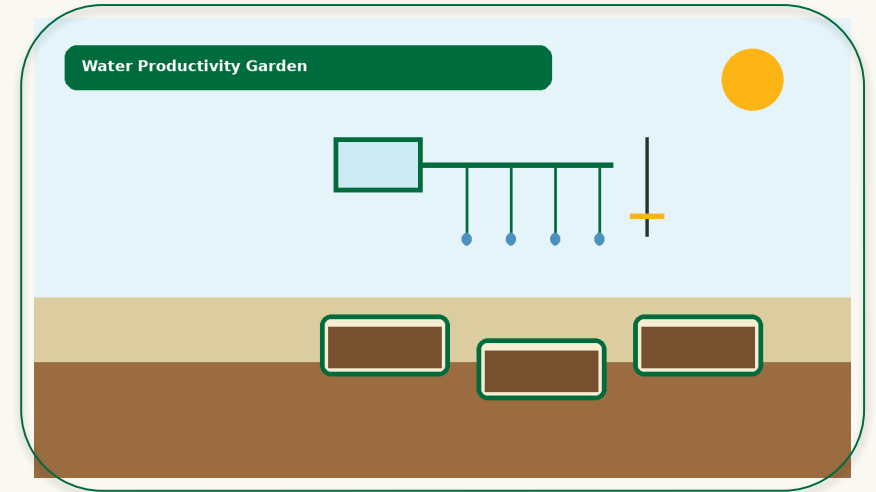


Water Productivity Garden

Measuring Crop Output per Water Input

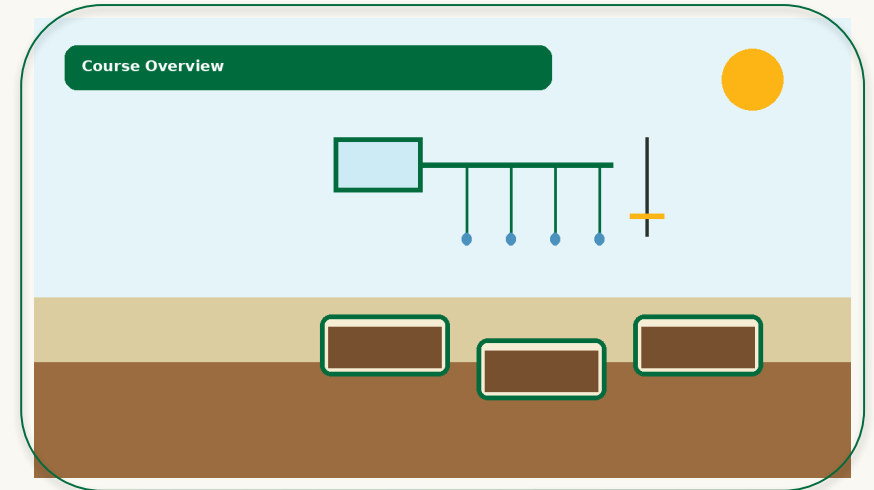
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide basic quantitative analysis; distinguish measured evidence from inference; connect results to management and engineering decisions.
- Core materials: teacher guide, system/plot drawings, crop and experiment plans, approved field tools, irrigation equipment, measuring tapes, scales, flow/pressure/moisture instruments where relevant, sensors/data sheets, clean harvest/sample containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit a production plan, engineering design, experimental protocol or analytical report appropriate to the course.



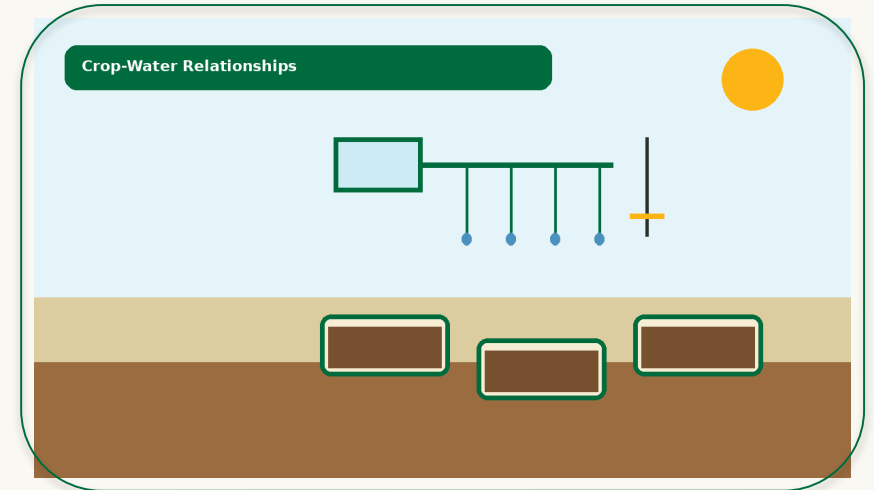
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Crop-Water Relationships

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain how crop growth depends on water availability, crop stage, rooting conditions and environmental demand.
- Distinguish water requirement, irrigation application and actual crop response.
- Recognize that both deficit and excess water can reduce performance.
- Use measured soil and crop observations to support decisions.
- Avoid assuming that lower irrigation always means greater efficiency.



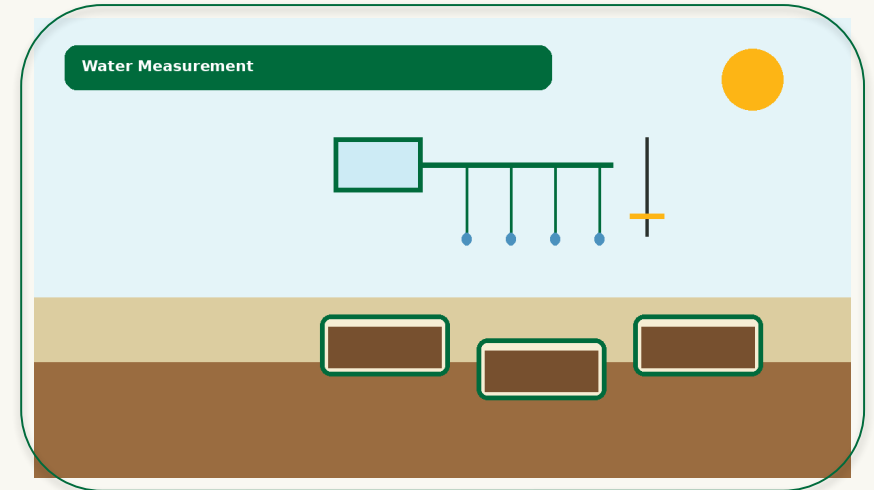
TEACHER PRACTICE

- Teacher task: map water input to root-zone condition and crop response.

Water Measurement

Teacher technical knowledge, laboratory/field practice and student supervision

- Measure irrigation volume using flow meters, calibrated containers, tank-level change or known discharge-duration relationships.
- Use consistent units and document whether values are measured or estimated.
- Include rainfall where reliable measurements are available and relevant to the water balance.
- Check equipment and calculations.
- Record water by treatment and date.



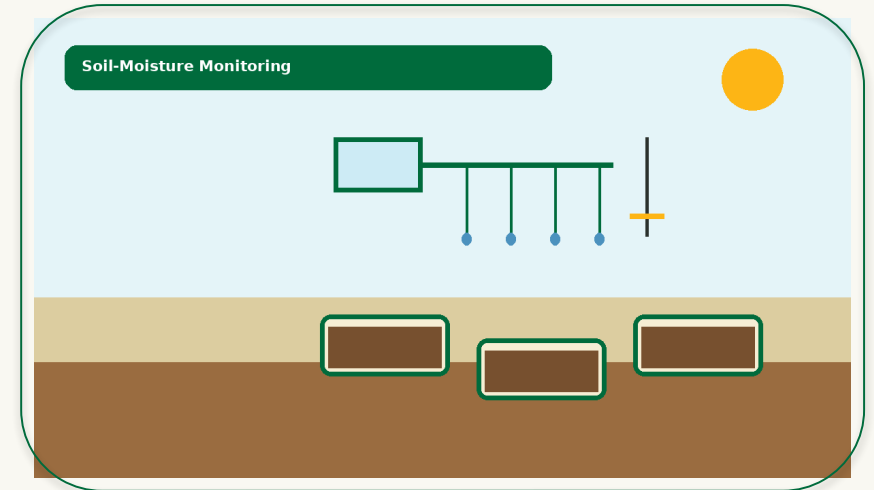
TEACHER PRACTICE

- Create a treatment-level water-input ledger.

Soil-Moisture Monitoring

Teacher technical knowledge, laboratory/field practice and student supervision

- Use sensors and/or standardized manual measurements at representative depths and positions.
- Measure at consistent times relative to irrigation.
- Use multiple locations to capture spatial variability.
- Record sensor units and calibration/context information.
- Interpret moisture trends alongside water inputs and crop stage.



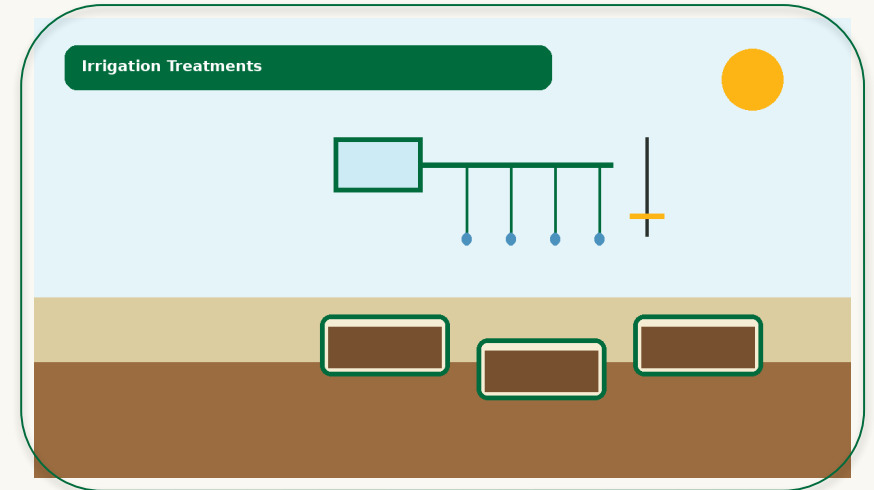
TEACHER PRACTICE

- Develop a soil-moisture monitoring protocol.

Irrigation Treatments

Teacher technical knowledge, laboratory/field practice and student supervision

- Define safe treatment levels based on approved management ranges, not severe damaging stress.
- Use a reference treatment and one or more alternative water applications/schedules.
- Replicate and randomize treatments where feasible.
- Keep crop, soil, fertility and measurement procedures consistent.
- Measure actual delivered water rather than relying only on intended treatment.



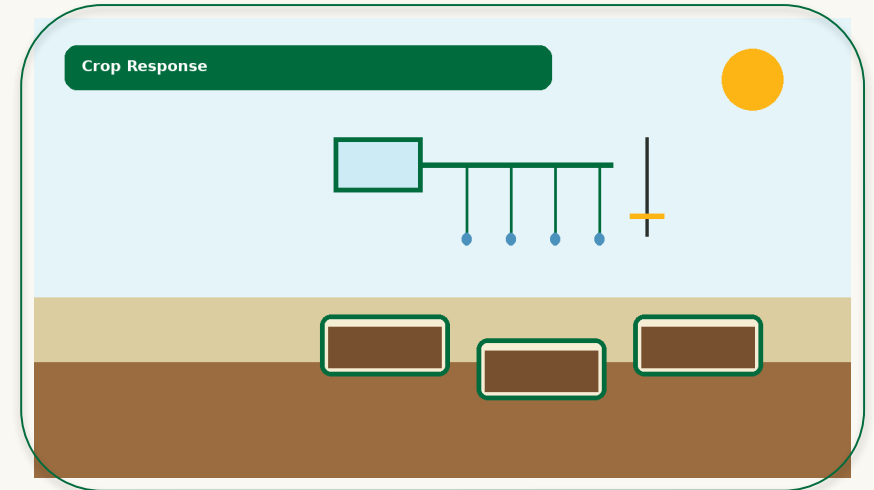
TEACHER PRACTICE

- Design a replicated irrigation-treatment experiment.

Crop Response

Teacher technical knowledge, laboratory/field practice and student supervision

- Measure establishment, growth, canopy, phenology, stress indicators and/or other relevant traits.
- Use consistent sampling across treatments.
- Track response over time to identify when differences emerge.
- Record pest, nutrient or other factors that may affect response.
- Avoid attributing all crop differences to water without considering confounders.



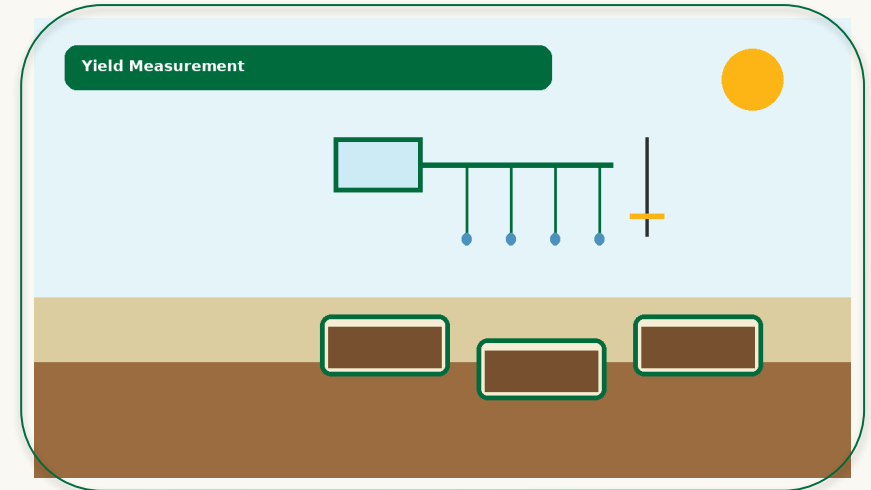
TEACHER PRACTICE

- Select four crop-response indicators and define measurement methods.

Yield Measurement

Teacher technical knowledge, laboratory/field practice and student supervision

- Harvest defined areas using common maturity criteria.
- Measure total and marketable yield where relevant.
- Record units, harvest date and treatment.
- Adjust comparisons for plot area and plant population when necessary.
- Preserve replicate-level yield data.



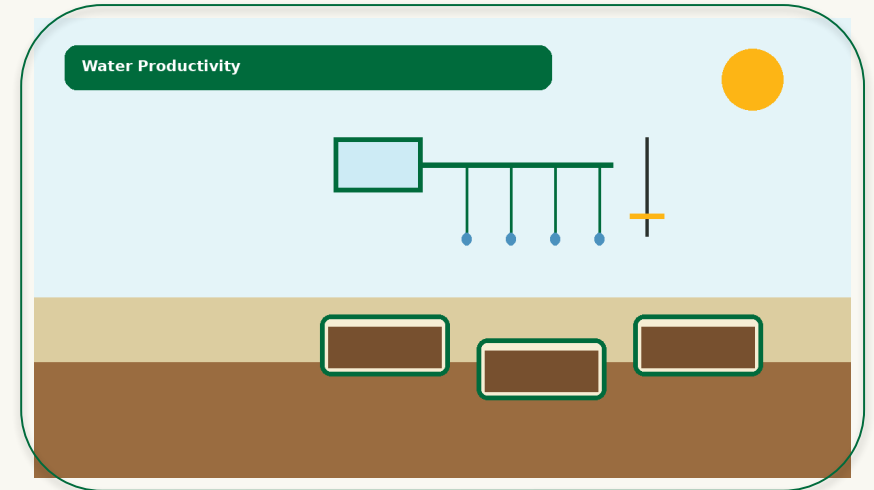
TEACHER PRACTICE

- Prepare a treatment-by-replicate yield sheet.

Water Productivity

Teacher technical knowledge, laboratory/field practice and student supervision

- Define water productivity as useful crop output divided by a specified water input, with numerator, denominator and units clearly stated.
- Use measured yield and water data from the same treatment boundary and period.
- Calculate per treatment and retain replicate information where possible.
- State whether the denominator is irrigation water only or a broader water input.
- Avoid comparing values calculated with different definitions.



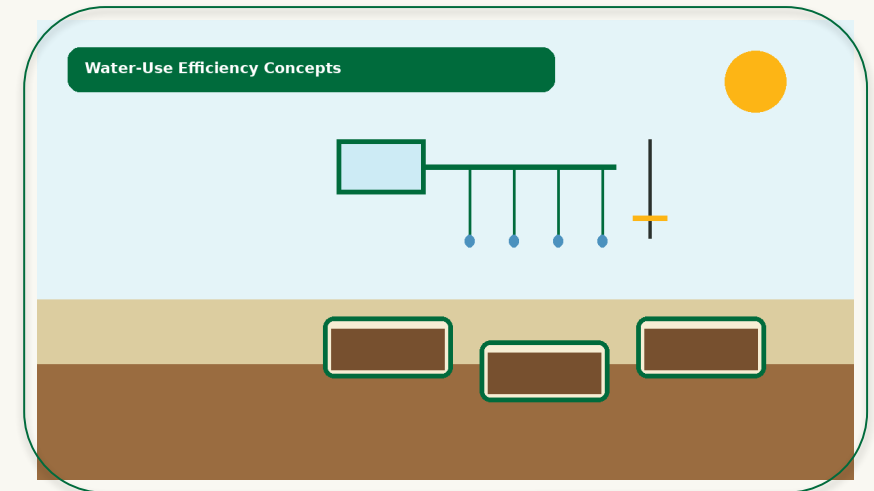
TEACHER PRACTICE

- Calculate water productivity for sample treatments.

Water-Use Efficiency Concepts

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain that 'water-use efficiency' can have different meanings in agronomy and plant physiology.
- Distinguish field-level yield per water input from physiological biomass/carbon gain per evapotranspiration or transpiration concepts.
- Use the term only after defining the numerator and denominator.
- Do not label irrigation water productivity as physiological WUE.
- Match the metric to the research question and available measurements.



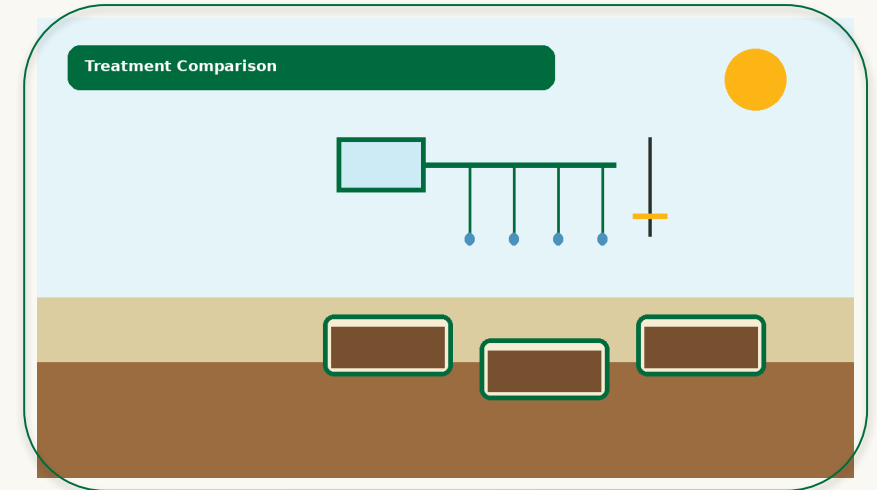
TEACHER PRACTICE

- Prepare a metric-definition table: irrigation WP versus physiological WUE.

Treatment Comparison

Teacher technical knowledge, laboratory/field practice and student supervision

- Compare water applied, soil-moisture behavior, crop response, yield and calculated water productivity.
- Use graphs showing both water input and crop output.
- Consider variability among replicates.
- Identify whether a treatment saves water at an unacceptable yield cost or maintains output with less input.
- Avoid choosing a 'best' treatment from one metric alone.



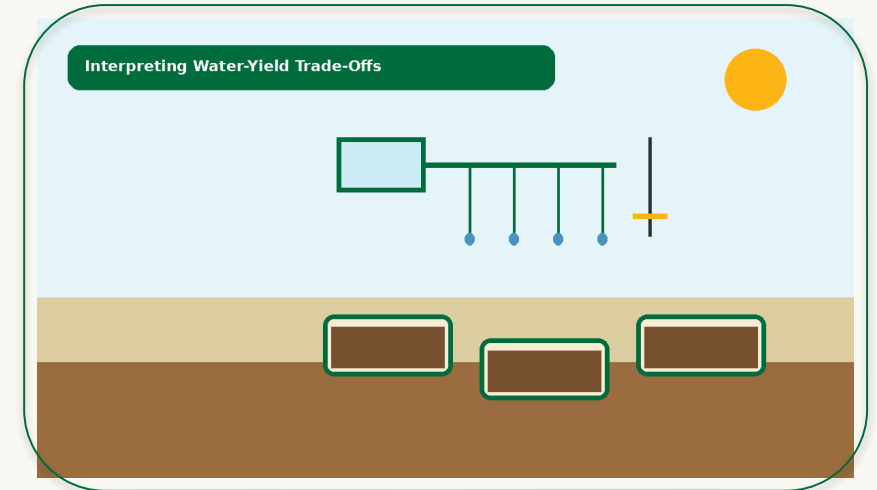
TEACHER PRACTICE

- Prepare a multi-criteria treatment comparison.

Interpreting Water-Yield Trade-Offs

Teacher technical knowledge, laboratory/field practice and student supervision

- Plot or tabulate yield against water input to visualize the production trade-off.
- Identify diminishing returns where supported by the data.
- Discuss management objectives: yield, water conservation, reliability or combined performance.
- Recognize uncertainty and site/season dependence.
- Recommend a management range rather than false precision when evidence is limited.



TEACHER PRACTICE

- Completion task: write an evidence-based water-yield management recommendation.